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# Polyester (PET) Degradation in Mild-Alkaline Solutions Assisted by Ultrasonication and UV-Activated Metal Oxides

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## Abstract

The exponential and continuous growth in the production and consumption of plastics, particularly polyethylene terephthalate (PET), due to their versatile applicability, has led to the accumulation of persistent plastic waste in the environment. This has created an urgent need to develop more sustainable and efficient recycling technologies. Although several recycling methods, such as mechanical and chemical recycling, have been implemented on an industrial scale, existing technologies still do not fully recover PET to its original quality or rely heavily on hazardous chemicals at elevated operating conditions. In this study, we explored a greener approach for PET depolymerization through hydrolysis under milder conditions, enhanced by mechanochemical degradation via ultrasonication, and integrated with UV-assisted treatment using five heterogeneous catalysts (ZnO, $ZnFe_2O_4$, $Fe_2O_3$, $Y_2O_3$, and SrO). The results of this study proved the significance of the mechanochemical effect of ultrasonication, which enhanced PET degradation by promoting chain scission, achieving up to 21.5% PET conversion and 19.0% terephthalic acid (TPA) yield in 1M NaOH hydrolysis. Among the catalysts tested under 0.1M NaOH hydrolysis, SrO exhibited the highest catalytic performance due to the formation of Sr(OH)2, which provides additional hydroxide ions for ester bond cleavage. However, excessive SrO loading led to $SrSO_4$ precipitation during acidification, requiring additional purification of the TPA product.

## Keywords

Polyester, Polyethylene terephthalate, Ultrasonication, Catalysis, Photocatalysis

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## Abstract

The exponential and continuous growth in the production and consumption of plastics, particularly polyethylene terephthalate (PET), due to its versatile applicability, has led to the accumulation of persistent plastic waste in the environment. This has created an urgent need to develop more sustainable and efficient recycling technologies. Although several recycling methods, such as mechanical and chemical recycling, have been implemented on an industrial scale, existing technologies still do not fully recover PET to its original quality or rely heavily on hazardous chemicals at elevated operating conditions. In this study, we explored a greener approach for PET depolymerization through hydrolysis under milder conditions, enhanced by mechanochemical degradation via ultrasonication, and integrated with UV-assisted treatment using five heterogeneous catalysts (ZnO, $ZnFe_2O_4$, $Fe_2O_3$, $Y_2O_3$, and SrO). The results of this study proved the significance of the mechanochemical effect of ultrasonication, which enhanced PET degradation by promoting chain scission, achieving up to 21.5% PET conversion and 19.0% terephthalic acid (TPA) yield in 1M NaOH hydrolysis. Among the catalysts tested under 0.1M NaOH hydrolysis, SrO exhibited the highest catalytic performance due to the formation of $Sr(OH)_2$, which provides additional hydroxide ions for ester bond cleavage. However, excessive SrO loading led to $SrSO_4$ precipitation during acidification, requiring additional purification of the TPA product.

## 1. Introduction

As the world population continues to grow, there is a corresponding increase in the production and consumption of plastics due to their unique material properties. Synthetic polymers or plastics are lightweight, durable, cost-effective, and easily manufactured [1, 2] making them popular in both consumer and industrial applications. However, worldwide consumers lack awareness of the negative consequences of discarding plastic waste. Over time, the accumulation of plastic waste in the environment has become a significant issue, particularly in countries with inadequate waste management systems. Currently, plastic waste management primarily relies on landfills [1, 3-5]. However, plastics are non-biodegradable and persist in landfills for hundreds of years. As the term "Forever Chemicals" implies, instead of degrading completely, they break down into microplastic fragments, which pose even greater challenges to manage due to the negative effects of plastic pollution in soil and marine environments, harming animal and human health [6]. Another common method of managing plastic waste is incineration, which is an easy option but with the downside of releasing greenhouse gases (GHG) among other toxic gases. Most plastic materials are derived from fossil fuels, which further releases significant amounts of GHG [7, 8], contributing directly to global warming and climate change. To address these issues, global research must focus more on sustainable plastic recycling processes as an alternative to minimize environmental impacts and to enhance the value of the plastics industry in a circular economy.

The global production by Plastics Europe (2023) reported that in 2023 413 million metric tonnes were produced, and only 36.5 million metric tonnes were recycled [9]. This shows that 91.2% of plastic produced ends up as plastic waste in landfills or aquatic environments. Among the various plastic types, polyethylene terephthalate (PET), which can be categorized as a thermoplastic type, is the most frequently used and commonly found in everyday life items such as in single-use food/beverage packaging, clothing/textiles, films, electronics, and automotive parts due to its excellent thermal and mechanical stability, good chemical resistance, and low water absorption [1, 7, 10, 11]. Reportedly, PET accounts for about 6.2% of the total plastic waste [9]. The current existing PET recycling technologies can be broadly categorized into two major methods for plastic degradation, known as mechanical and chemical recycling. Mechanical recycling is the most widely and commonly used method due to its simplicity, low-cost, and environmental friendliness. The mechanical recycling process involves sorting, size reduction (chopping or milling), and heat treatment to melt and reform plastic waste into pellet form [10, 12]. However, mechanical processing downgrades the PET's original physical properties by lowering its molecular weight, which reduces the melt viscosity. Contamination during mechanical processing causes discoloration. In addition, the presence of water and acid during the heat treatment, result in random chain-scission from the building block of polymer [2, 13].

In contrast with mechanical recycling, chemical recycling breaks PET down into its monomer followed by polymerization through traditional processes. Thus, PET waste can be re-converted into the original PET without quality degradation [2]. Chemical recycling of PET involves cleavage of its ester bonds, which can be achieved using one of the following solvent-based processes: aminolysis, ammonolysis, alcoholysis, glycolysis, and hydrolysis. Recent studies have shown that hydrolysis is the most facile method for PET depolymerization to terephthalic acid (TPA) and ethylene glycol (EG), which are the basic raw monomers of PET compared to its alternative monomers [12, 14, 15].

PET hydrolysis is a chemical reaction that occurs in the presence of water, breaking down polymer chains. It consists of three main pathways: neutral hydrolysis, acidic hydrolysis, and alkaline hydrolysis. Depolymerization in natural hydrolysis can occur naturally over time, but it is much slower and requires a very high temperature (250-300 ˚C) and pressures (1.5-4 MPa). Acidic and alkaline hydrolysis occur more rapidly around 3 to 5 h at temperatures of 210-250 ˚C under 1.4-2 MPa [16]. Common acid and base catalysts are sulfuric acid, nitric acid, and sodium hydroxide [4, 10]. In addition to using strong acid and alkaline solutions as raw materials in the PET recycling process, opposing acids/bases are required to neutralize and separate out the TPA monomer, leading to tremendous environmental impacts because the acids/bases cannot be recycled/reused. These disadvantages, combined with the high operating and capital costs required to maintain elevated temperatures and pressures, make PET hydrolysis an unattractive option for use in the PET recycling industry.

To address the drawbacks of the hydrolysis process, researchers have been exploring alternative options for reducing its costs and environmental impact. One option is to use a heterogeneous (solid) catalyst. Unlike homogeneous (e.g., liquid) catalysts, solid catalysts can be easily separated from the resulting liquid mixture after the reaction, for example by filtration or centrifugation [17]. Kang et al. demonstrated the use of ZSM-5 zeolite with microwave-assisted hydrolysis, achieving up to ~100% yield of TPA in 30 minutes at a reaction temperature of 230 ˚C and 30 bars of pressure [14]. Although this approach eliminates the use of sodium caustic/acidic solutions, the energy requirement remains significant. The study explored other catalysts such as ZSM-5 with varying Si/Al to control the acidity of the catalysts (increasing the number of Brønsted acid sites) to enhance the hydrolysis efficiency. In addition, the recyclability of the heterogeneous catalysts was demonstrated, with high catalyst activity maintained even after six cycles of use compared to a fresh catalyst [14]. Similarly, Pereira et al. studied various acid catalysts, including zeolite, inorganic acids, and metal salts for PET hydrolysis. Their research highlighted that heterogeneous zeolite catalysts performed poorly at a low reaction temperature of 200 ˚C, with HY and HZSM-5 zeolites achieving only 18% and 10% TPA yields, respectively. However, increasing the temperature up to 270 ˚C significantly increased TPA yield, with HY achieving up to 85% in 30 minutes, demonstrating the potential of heterogeneous catalysts [18]. Liguouri et al., on the other hand, achieved 100% conversion of PET using a ZnO catalyst at 180°C [3]. Several heterogeneous catalysts have attracted great interest due to their excellent catalytic efficiency in glycolysis such as sulfated oxides, iron oxide [19], yttrium Oxide ($Y_2O_3$) [20], single metal oxides ($Mn_3O_4$, $Co_3O_4$), and mixed metal oxides ($ZnMn_2O_4$, $CoMn_2O_4$, $ZnFe_2O_4$, $ZnCo_2O_4$, and $ZnFe_2O_4$) [21].

To further improve the degradation of PET at milder temperatures and pressure, additional mechanical forces have been integrated. This is because it is well-established that mechanical forces can trigger and accelerate the degradation of ester bonds in polymers via promoting chain scission [22]. Techniques such as extension, compression, swelling, ball-milling [23, 24], and ultrasonication [22, 23] have shown promising results in accelerating the breakdown of long polymer chains when combined with chemical hydrolysis. Zhuang et al. implemented mechanical force by using ball-milling and ultrasonication to accelerate the degradation of vinyl polymers via the hydrolysis process, with results indicating that mechanical forces improve the degradability of the polymer material [23]. Similarly, Paliwal et al. evaluated the ultrasound-assisted alkaline hydrolysis of PET using a 10% NaOH (w/w) solutions and the catalyst Tetrabutyl Ammonium Iodide (TBAI) at a constant reaction temperature of 90 ˚C and atmospheric pressure [25]. With ultrasonication, they reached a 100% PET conversion in 45 min as opposed to 65 minutes without, demonstrating the effectiveness of incorporating ultrasonication; however, the recoverability of the catalyst would be challenging [25].

Other researchers have investigated the use of new heating sources such as microwave-assisted processes [1, 26, 27], or UV light [28] to accelerate the PET degradation rates, promising enhancements for PET hydrolysis under milder conditions. Another approach is to use a photocatalyst, which is expected to absorb energy from the UV light, generating electron-hole pairs, facilitating chain cleavage in the ester bond, allowing the reaction to take place at milder conditions [5, 29]. However, only a few studies have investigated heterogeneous photocatalysts with PET hydrolysis.

The objective of this work was to evaluate the ability of common metal oxides to assist (whether photocatalytically or not) the hydrolysis of PET. Ultrasonication was also adopted to assist with the initial breakdown of PET to make the material more available for catalytic degradation. The intent of this study was to develop a novel UV- and ultrasonication-assisted mild alkaline PET hydrolysis process that can operate at atmospheric pressures with temperatures below 100°C. The selected catalyst materials consisted of commercial ZnO, $ZnFe_2O_4$, $Fe_2O_3$, $Y_2O_3$, and SrO based on observed performances within literature as either thermal or photocatalysts [30], material availability, and applicability to future catalyst development. ZnO, particularly, was selected based on its ability to assist in the hydrolysis of polyesters, as demonstrated by Liguori et al. [3] and its photocatalytic behaviour in degrading pharmaceuticals and azo dyes [31, 32].

## 2. Experimental

### 2.1 Materials

The five metal oxide catalysts, zinc oxide (ZnO), zinc iron oxide ($ZnFe_2O_4$), yttrium oxide ($Y_2O_3$), Iron (III) oxide ($Fe_2O_3$), and strontium oxide (SrO), which were purchased from Sigma Aldrich, VWR, and Fisher Scientific. Sodium hydroxide (≥98%, NaOH) was purchased from Fisher Scientific. PET films were obtained from McMaster-Carr. The PET film was cut into small flakes of approximately 0.4x0.8 $cm^2$. A thermogravimetric analysis of the PET film (10 °C/min to 900°C under $N_2$ at 50 mL/min) revealed that the film closely matched that of pure PET reported in literature [33, 34], with no substantial impurities present (see Figure S2).

### 2.2 Neutral Hydrolysis Process

For all experiments, 5 g of PET flakes were added to a flat-bottom quartz tube, followed by the addition of 170 mL of deionized water (sufficient to fully cover the top of the of the ultrasonication probe, immersed about 1.25 cm). The neutral hydrolysis of PET was conducted inside a sound-insulated box that consisted of an ultrasonic dismembrator probe (Banson US) (referred to as the US probe), six UV grow light panels (10 W each, total power 60 W) with full-spectrum coverage, and a magnetic stirrer. Additionally, to avoid the operating limitations of US system failure due to t heat generation, a cooling system was installed in three positions: a cooling water coil to cover the head of US converter, a fan directed at the extension and the US probe, and a refrigeration system along the side walls of the box as an overall cooling system.

Once the reaction mixtures were set up properly in position, an airline from an air pump was placed into the tube to promote mixing and oxidation reactions. A thermocouple was also inserted in the reactor to continuously monitor the reaction temperature. Before starting the reaction, the cooling systems were turned on, and the US amplitude was adjusted and maintained for 2 hours. 10 mL of DI water was added every 2 hours to keep the solution level at the same position. After the reaction was completed, the mixture was allowed to cool to room temperature inside the insulated box. Unreacted PET (UnPET) was first separated using a sieve, while the remaining microplastics and solution mixture were subjected to vacuum filtration. Ethylene glycol mixture was recovered from the filtrate, while the white particles of terephthalic acid (TPA), and microplastics were collected on a 0.2 µm glass fiber filter paper. The UnPET and the filter paper were then dried in an oven at 110°C for 12 hours. To ensure that there were no microplastics contaminated with TPA, another purification step was introduced by dissolving the paper filter in DMSO, which allows only TPA to dissolve, followed by a second filtration step to completely separate the microplastics from the TPA. Then, UnPET was weighed to determine the amount of unreacted part and further calculate the PET conversion percentage as shown in Eq. (1):

$$\%\ \text{PET Conversion} = \left(\frac{W_i - W_f}{W_i}\right) \text{x } 100\% \qquad (1)$$

Where $W_i$ is the initial weight of PET before the reaction, and $W_f$ is the weight of unreacted PET after the reaction and sieve separation. All reported values in the following sections of this report are based on the average of two independently repeated experiments to ensure accuracy and reliability. In addition, the TPA concentration in DMSO was measured by using UV-VIS spectroscopy, with quantification based on a prepared TPA standard calibration curve.

### 2.3 Sodium Hydroxide (NaOH) Hydrolysis Process

The NaOH hydrolysis process followed the same general procedure as the neutral hydrolysis, but instead of deionized water, 170 mL of 1 M NaOH solution was added to the quartz tube. The NaOH hydrolysis reaction occurs as shown in **Figure 1**.

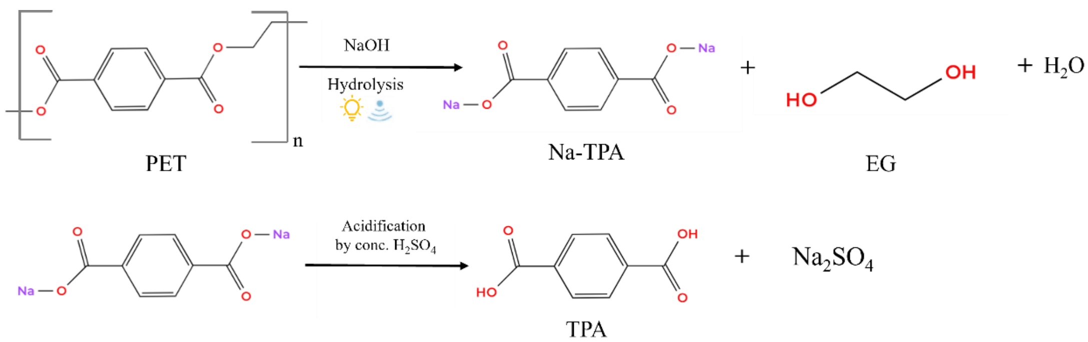



**Figure 1:** NaOH hydrolysis process of PET

After the hydrolysis reaction, the UnPET was rinsed multiple times with DI to remove any residual NaOH attached to the surface of the UnPET. Then, the solution, which primarily contained disodium terephthalate (Na-TPA), was recovered to pure TPA by an acidification step using concentrated sulfuric acid ($H_2SO_4$). This step was performed by slowly adding $H_2SO_4$ dropwise into the mixture solution under constant stirring in a full ice bucket to control the heat generated during the acidification reaction. The addition of acid continued until the pH of the solution reached 1-2 to ensure the complete precipitation of TPA from its disodium salt form. The white precipitate of TPA was then collected by vacuum filtration and washed thoroughly with deionized water to remove any remaining sodium sulfate or unreacted acid until the pH reached a neutral value. The product on the filter paper was then dried in an oven at 110°C for 12 hours. These steps were repeated with 0.1 M NaOH and five catalysts (ZnO,

$ZnFe_2O_4$, $Y_2O_3$, $Fe_2O_3$, and SrO). The concentration of catalyst evaluated were 1%, 2.5%, and 5% weight of PET. After the sieve separation of UnPET, the catalysts were recovered via centrifugation until the mixture solution became clear, followed by vacuum filtration. The catalysts separated in the conical glass centrifugation tubes were dried in an oven overnight to eliminate moisture. Afterwards, the catalyst powders were collected and placed in ceramic crucibles for calcination at 600 °C to remove any residual UnPET (micro-sized). The percentage of TPA yield was calculated based on the moles of the repeating unit basis by the following Eq.2:

$$\%\text{Yield of TPA} = \frac{\textit{Moles of TPA, produced}}{\textit{Moles of TPA, theoretical}} x\ 100\% \qquad (2)$$

where Moles of TPA, produced is the moles of TPA produced through the reaction, and Moles of TPA, theoretical is refers to the theoretical moles of TPA that could have been produced based on the assumption of 100% PET conversion. The surface morphology of PET films before and after the hydrolysis reaction were observed under a microscope using an Olympus GX41 optical microscopy to determine surface changes on the PET film due to the physical and chemical mechanisms.

## 3. Results and Discussions

### 3.1 Neutral Hydrolysis Reaction

In the initial phase of this study, a neutral hydrolysis reaction was conducted to evaluate the effect of ultrasonication, UV light, and the presence of heterogeneous catalysts on the degradation of PET. The experimental results demonstrated an insignificant PET conversion. As a result, terephthalic acid (TPA), the main product of PET hydrolysis, was either not visibly formed or only detected in trace amounts. This made it difficult to confirm any degree of depolymerization because the amount of TPA could not be directly weighed after the reaction. To address this issue, DMSO was introduced as a solvent due to its high solubility performance [35], enabling the dissolution of any traces of TPA via UV-VIS. However, DMSO showed a complex absorbance background causing interference with the UV-VIS readings that led to an overestimation of TPA concentration. Therefore, the focus of this study shifted to low-concentration NaOH hydrolysis aiming to develop an environmentally friendly hydrolysis process, while optimizing the reaction conditions to improve the PET recycling performance.

## 3.2 Mild Alkaline PET Hydrolysis Reaction

### 3.2.1 Effect of Ultrasonication Amplitude

To investigate the effects of ultrasonication amplitude (e.g., ultrasonic intensity) on the degradation of PET films. Three different amplitude levels: 3 (48.5 µm), 5 (75 µm), and 7 (103.6 µm) were evaluated during the alkaline hydrolysis of PET using 1M NaOH concentration. As shown in **Figure 2**, both PET conversion and TPA yield increased with increasing ultrasonic amplitude. PET conversion increased from 3.07% at amplitude 3 to 21.47% at amplitude 7 (103.6 µm), while the TPA yield increased from 2.50% to 18.98%, as shown in Table 1. The increased TPA yield is explained by the mechanochemical effects induced by ultrasonication during the hydrolysis reaction. These experimental observations agree with previous research [22, 23, 25]. The observed increase in reaction temperature at higher ultrasonication amplitudes agrees with the fundamental principles of ultrasonication, which is caused by the cavitation phenomenon generated during ultrasonication. Ultrasonic cavitation creates microbubbles that continuously collapse due to compression and expansion phases, leading to the generation of shock waves and shear forces on the PET surface; which further enhances the continuous stretching and/or elongation of the polymer chains promoting chain scission [24, 28].

To confirm the contribution of ultrasonication beyond the thermal effects, a control experiment was conducted with a heating band and temperature controller instead of ultrasonication. The temperature was maintained at 85 °C – the median temperature observed in the previous experimental run using 1 M NaOH solution at an ultrasonic amplitude setting of 7 (see **Figure 3**). Interestingly, under these conditions, PET conversion in the absence of ultrasonication was lower than that in the ultrasonically assisted reactions. These experimental observations confirm the synergistic effect of temperature and ultrasonication (e.g., mechanical effect). To further support this observation, optical microscopy was employed to investigate the surface morphology of PET films before and after the hydrolysis reaction. As shown in **Figure 4**(a), the initial PET films before the reaction exhibited a smooth and uniform surface which contrasted with UnPET film of 1M NaOH under ultrasonication; which displayed severe surface damage, characterized by two distinct features: chemical erosion caused by NaOH, observed as circular etching patterns, and physical scratches and cracks induced by the mechanical effects of ultrasonication. This dual effect on the surface of the UnPET is commonly referred to as "mechanochemical" degradation.

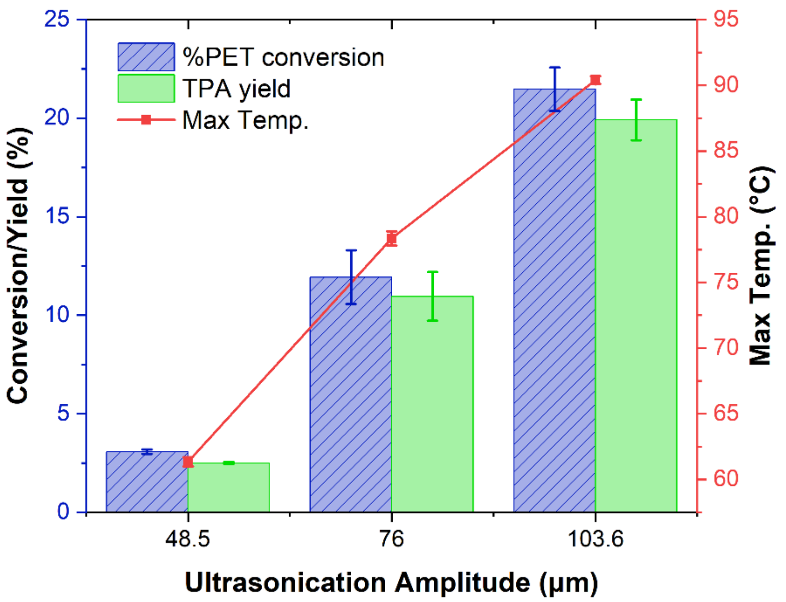
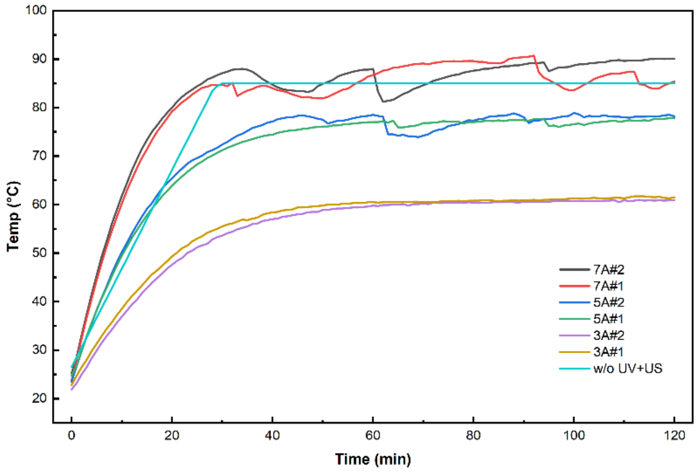



**Figure 2:** Effect of ultrasonication amplitude on PET conversion, TPA yield, and maximum temperature with 1M NaOH.

**Table 1:** % PET Conversion, TPA yield, and standard errors of 1M NaOH hydrolysis run.

| Run | PET Conversion (%) | TPA Yield (%) |
|---|---|---|
| NaOH 1M@3A | 3.07± 0.11 | 2.50± 0.05 |
| NaOH 1M@5A | 11.93± 1.36 | 10.96± 1.24 |
| NaOH 1M@7A | 21.47± 1.10 | 18.98± 1.11 |
| NaOH 1M w/o UV-US 85℃ | 17.72± 0.47 | 17.07± 0.04 |




260 **Figure 3:** Temperature profiles of various experiments with 1 M NaOH with and without (w/o) UV-US assistance at different amplitude levels: 3 (“3A” – 48.5 µm), 5 (“5A” – 75 µm), and 7 (“7A” – 103.6 µm). Temperature drops coincide with DI water additions.

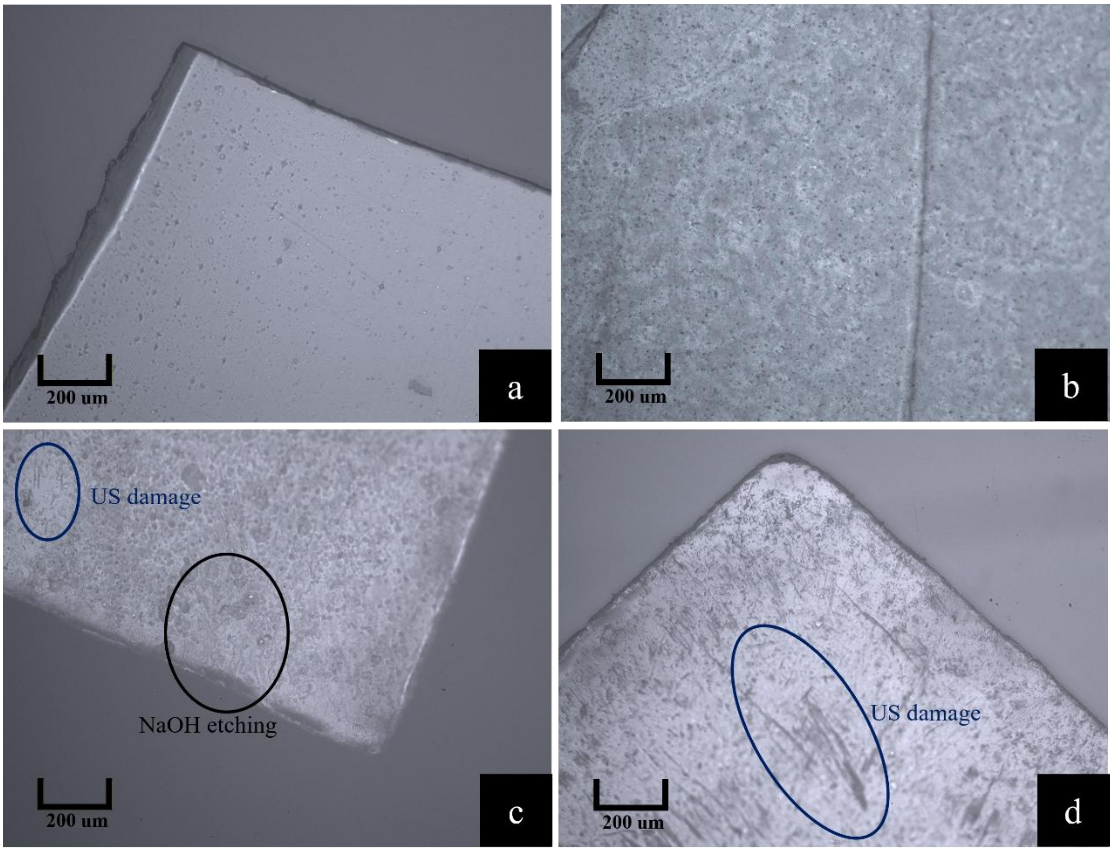



**Figure 4:** Micro pictures of PET Film Surface at 10X magnification. a) Initial PET film b)
265 UnPET of 1M NaOH without UV-US at 85 ˚C c) UnPET of 1M NaOH @7A d) UnPET of 0.1M NaOH @7A

### 3.2.2 Effect of Heterogeneous Catalysts

It is well-established that the concentration of NaOH plays a crucial role in PET alkaline hydrolysis reactions. A high concentration of NaOH provides more hydroxide ions ($OH^-$), which act as nucleophiles that attack the carboxyl ester group (-COO-R) in the PET backbone, thereby accelerating the depolymerization reaction. This trend has been reported in numerous studies [36-38] and was confirmed in this project, where % PET conversion dropped significantly from 21.47% to 1.49% when the concentration of NaOH was reduced from 1M to 0.1 M. However, using highly concentrated alkali solutions poses challenges in terms of cost, safety, and environmental impact, particularly when considering large-scale applications. To address this limitation, this study explores the feasibility of operating at a lower NaOH concentration of 0.1 M. Five heterogeneous catalysts —$ZnFe_2O_4$, $Y_2O_3$, ZnO, $Fe_2O_3$, and SrO —commonly studied and widely applied in PET recycling, especially in glycolysis, were investigated to enhance the depolymerization process. The depolymerization of PET was evaluated for each of these heterogeneous catalysts under ultrasonication at an amplitude of 103.6 μm and UV light. Among these, SrO demonstrated the highest catalytic activity, achieving a PET conversion of 1.77% and a TPA yield of 0.61%, outperforming all other candidates as shown in Table 2. The significant increase in both PET and TPA conversion can be explained by the reaction of SrO with water, which forms strontium hydroxide ($Sr(OH)_2$) [39, 40] as shown in the reaction equation below.

$$SrO_{(s)} + H_2O_{(l)} \leftrightarrow Sr(OH)_{2(s)} \leftrightarrow Sr^{2+}_{(aq)} + 2OH^-_{(aq)} \quad (3)$$

This strong base provides additional hydroxide ions, further enhancing the nucleophilic attack on the ester bonds. In contrast, the other catalysts $ZnFe_2O_4$, $Y_2O_3$, ZnO, and $Fe_2O_3$ did not significantly improve the hydrolysis performance compared to the reaction without a catalyst, which did not meet our initial expectations. These four catalysts were selected based on their reported ability to provide Lewis or Brønsted acidic sites, which could potentially induce a partial positive charge on the ester carbonyl group, thereby facilitating nucleophilic attack by water molecules [14, 19-21]. Moreover, these metal oxides are also well-known as photocatalysts, as confirmed by their light absorption capabilities in the UV–Vis region, as shown in Figure S1. Their absorption properties suggest potential for photocatalytic activation under light irradiation, which could, in theory, promote hydrolysis due to the formation of hydroxyl radicals (•OH) [5, 29] that are produced during the oxidation process in the photocatalytic mechanism, as shown below:

$$\text{Photocatalyst} + h\nu \rightarrow h^{+} + e^{-} \tag{4}$$
$$h^{+} + H_2O \rightarrow \cdot OH + H^{+} \tag{5}$$
$$h^{+} + OH^{-} \rightarrow \cdot OH \tag{6}$$
$$e^{-} + O_2 \rightarrow \cdot O_2^{-} \tag{7}$$
$$\cdot O_2^{-} + e^{-} + 2H^{+} \rightarrow H_2O_2 \tag{8}$$
$$e^{-} + H_2O_2 \rightarrow \cdot OH + OH^{-} \tag{9}$$
$$\text{PET} + \cdot OH \rightarrow \text{PET degradation} \tag{10}$$

However, under the conditions used in this study, photocatalytic enhancement or improvement by acidic site was not observed, which might be due to insufficient light intensity or the possibility that the active sites on these commercial oxides were insufficiently accessible to activate and cause chain scission to the ester functional groups.

**Table 2:** % PET Conversion, TPA yield, and standard errors of 0.1M NaOH hydrolysis run with five different heterogeneous catalysts.

| Run | PET Conversion (%) | TPA Yield (%) |
|---|---|---|
| NaOH 0.1M@7 | 1.49± 0.01 | 0.32± 0.12 |
| NaOH 0.1 M + $ZnFe_2O_4$ | 1.33± 0.21 | 0.43± 0.06 |
| NaOH 0.1M +$Yt_2O_3$ | 1.58± 0.10 | 0.36± 0.07 |
| NaOH 0.1M+ ZnO | 1.36± 0.01 | 0.00* |
| NaOH 0.1M + $Fe_2O_3$ | 1.01± 0.03 | 0.00* |
| NaOH 0.1M + 1% SrO | 1.77± 0.13 | 0.61± 0.02 |

* The amount of TPA was too low to measure using an analytical balance

### 3.2.3 Effect of Catalyst Loading

Based on its promising performance, SrO was selected for further investigation to evaluate the influence of catalyst loading on the depolymerization of PET. The SrO content was varied from 1 wt% to 5 wt% while maintaining the other reaction conditions constant. As shown in **Figure 5** and **Table 3Error! Reference source not found.**, both PET conversion and TPA yield increased with increased catalyst loading. At a low catalyst loading of 1 wt%, PET conversion reached only 1.77%, with a corresponding TPA yield of 0.61%, but when the SrO content was increased up to 2.5 wt%, there was no significant improvement in PET conversion. However, the TPA yield increased to 1.03%. This observation suggests that the excess of hydroxyl-free ions ($OH^-$) from the increased SrO loading may have promoted further breakdown of the PET intermediates into TPA. When the catalyst loading was further increased up to 5 wt%, much more $OH^-$ were available to attack both the ester bonds in the PET backbone and the remaining intermediates, leading to a substantial increase in PET conversion and TPA yield, reaching 4.89% and 1.54%, respectively. Although increasing catalyst loading would help increase PET conversion and TPA yield, it also resulted in a significantly higher concentration of $Sr^{2+}$ ions dissolved in the reaction mixture. Upon acidification with $H_2SO_4$, these $Sr^{2+}$ ions reacted with sulfate ions to form strontium sulfate ($SrSO_4$), which co-precipitated with TPA and caused contamination. To remove the $SrSO_4$ and purify the TPA, additional post-treatment steps are required such as converting the solution back into a salt solution (Na-TPA) using NaOH. $SrSO_4$ remained as an insoluble solid and could be separated from the purified Na-TPA. A final acidification step was then performed to recover purified TPA.

Formation of $SrSO_4$ suggests that a significant portion of SrO was converted into $Sr^{2+}$ and $OH^-$ ions rather than remaining in its original form. To further evaluate the effectiveness of SrO while minimizing $SrSO_4$ precipitation, NaOH concentration was increased to 0.34 M, which is the theoretical minimum NaOH concentration required to convert five grams of PET films completely. Accordingly, increasing the concentration of $OH^-$ ions should shift the equilibrium towards insoluble SrO, thereby reducing the dissolution of $Sr^{2+}$ ions into the solution. The results aligned well with this expectation, showing that at 0.1M NaOH with 5 wt% SrO, the amount of SrSO4 precipitated was 75.38 mg. However, when the NaOH concentration was increased to 0.34 M under the same catalyst loading, the amount of recovered $SrSO_4$ dropped significantly to only 2.83 mg. For comparison, the experiment using 0.34 M NaOH without SrO was also conducted, which showed that PET conversion and TPA yield were roughly half of those with SrO. This suggests that SrO assisted in PET chain scission by contributing hydroxyl radicals (·OH) produced via light absorption.

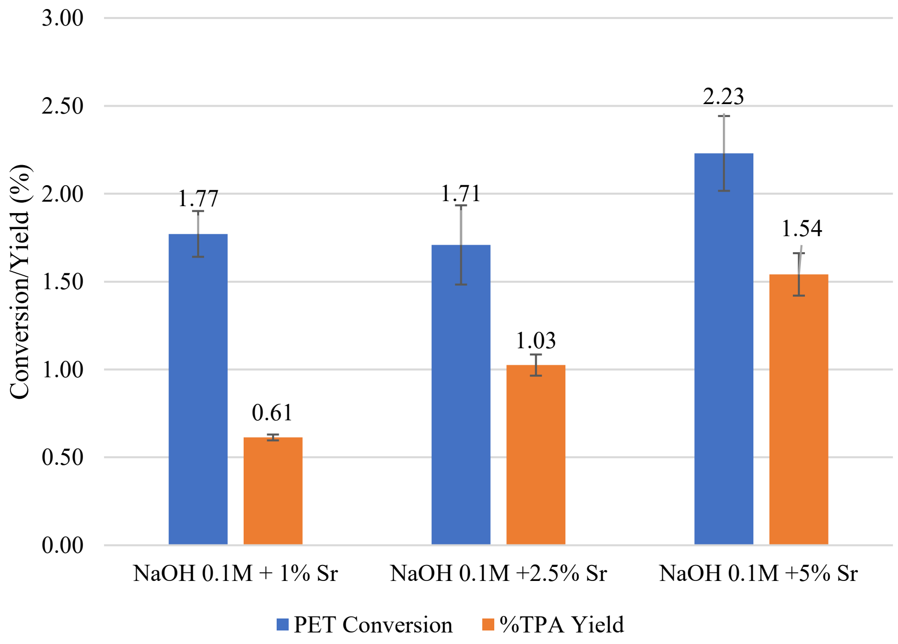



**Figure 5:** Effect of SrO catalyst loading on PET conversion and TPA yield

**Table 3:** % PET Conversion, TPA yield, standard errors, and amount of $SrSO_4$ formation of 0.1M and 0.34 M NaOH with SrO catalyst at various catalyst loadings.

| Run | PET Conversion (%) | TPA Yield (%) | Amount of $SrSO_4$ (mg) |
|---|---|---|---|
| NaOH 0.1M + 1% Sr | 1.77± 0.13 | 0.61± 0.02 | N/A |
| NaOH 0.1M +2.5% Sr | 1.71± 0.23 | 1.03± 0.06 | N/A |
| NaOH 0.1M +5% Sr | 2.23± 0.21 | 1.54± 0.12 | 75.38 |
| **NaOH 0.34M** | 2.86 ± 0.00 | 2.05±0.04 | - |
| NaOH 0.34M + 5% Sr | 5.61± 0.01 | 4.92± 0.13 | 2.83 |

### 3.2.4 Analysis of TPA Purity

Fourier-transform infrared (FTIR) spectroscopy was employed to confirm the chemical structure and assess the purity of TPA recovered after post-treatment. Following the reaction process described in Section 3.1.3, where increasing SrO catalyst loading led to the formation of $SrSO_4$ as a byproduct. The FTIR spectra of the recovered TPA and pure commercial TPA were compared to evaluate the structural similarity based on the absorption peaks corresponding to the characteristic functional groups, and the results are shown in **Figure 6**. The initially recovered TPA before the second post-treatment exhibited all the key absorption bands present in the pure TPA spectrum. A broad peak between 2500 and 3000 cm$^{-1}$ was attributed to O –H stretching from the carboxylic acid group, while a sharp peak near 1685 cm$^{-1}$ corresponded to C=O stretching of the carbonyl group, and a peak in the range of 1575-1425 cm$^{-1}$ is related to aromatic ring. However, in comparison to the commercial TPA, the recovered sample also displayed additional peaks characteristic of sulfate ions ($SO_4^{2-}$), observed in the range of 1200–900 cm$^{-1}$ and around 600 cm$^{-1}$ which confirmed the presence of $SrSO_4$. After applying the second post-treatment step no additional peak of $SO_4^{2}$ was detected, which confirms that $SrSO_4$ was effectively removed during the post-treatment step, resulting in a purified TPA product.

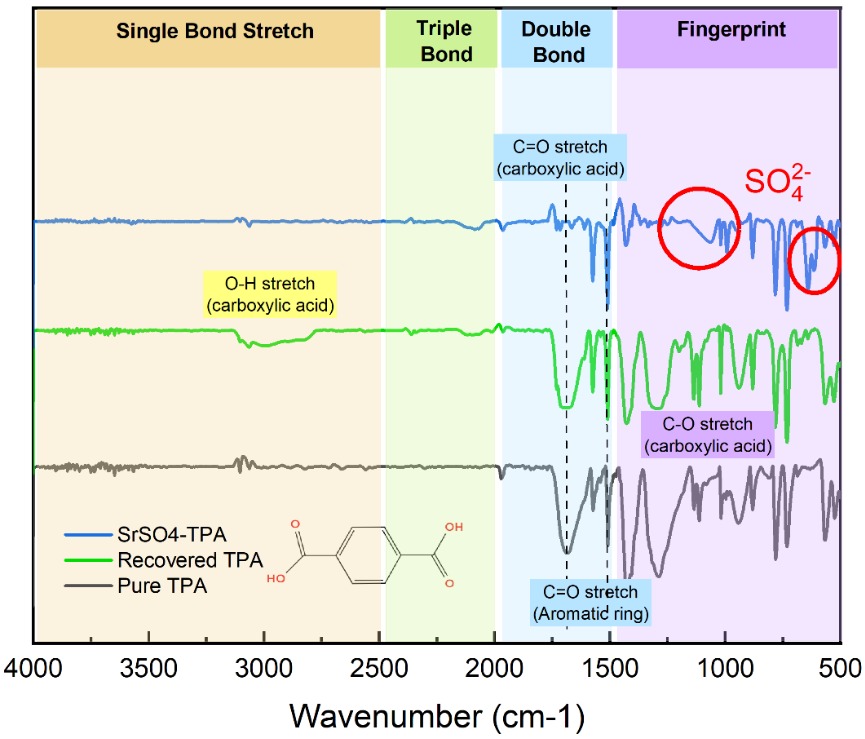



**Figure 6:** FTIR spectra of $SrSO_4$-TPA, separated TPA, and commercial (e.g., pure) TPA.

## 4. Conclusion

This study presents an approach to the catalytic recycling of polyethylene terephthalate (PET) under mild alkaline hydrolysis assisted by the mechanochemical action of ultrasonication coupled with UV-irradiation. This work also evaluated the performance of five heterogeneous catalysts (ZnO, $ZnFe_2O_4$, $Y_2O_3$, $Fe_2O_3$, and SrO) on PET depolymerization. Initial trials using neutral hydrolysis revealed low efficiency in TPA production, with difficulty in measuring TPA concentration due to the complexity of the DMSO background interference with absorbance in UV-VIS measurements. These limitations shifted the focus of the study to low-concentration alkaline hydrolysis using NaOH, while maintaining a recycling process that is environmentally friendly, and improving PET depolymerization.

The application of ultrasonication caused a significant improvement in PET degradation. Increasing the ultrasonication amplitude enhanced PET conversion and TPA yield. This improvement can be explained via the mechanochemical effects generated by cavitation, which produces intense localized heat and shear forces that physically disrupt the PET surface and promote the cleavage of ester bonds, thereby accelerating the depolymerization reaction. The catalyst screening was conducted under 0.1M NaOH hydrolysis. Among the five catalysts tested, SrO exhibited the highest catalytic performance, significantly improving the PET conversion and TPA yield due to the formation of $Sr(OH)_2$, which provides additional hydroxide ions for ester bond cleavage. However, higher SrO loadings introduced $SrSO_4$ precipitation during the acidification step, which requires the application of an additional post-treatment process to purify the TPA product.

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